

# TODD TAKALA

PRINCIPAL-LEVEL DATA SCIENTIST

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## PROFESSIONAL SUMMARY

Individual contributor holding the top IC grade at both Caterpillar and Ford. 25+ years in capital equipment engineering, from the shop floor as a mechanic and welder to enterprise machine learning. Ships production ML for predictive maintenance, forecasting, anomaly detection, and NLP built on failure data, IoT telemetry, and ERP systems, and owns the full MLOps lifecycle on AWS and Snowflake. Sets standards adopted beyond the immediate team, including Amazon's fleet design life standard and dealership-wide data governance at Empire Cat. Caterpillar CEO Award recipient and 2x Amazon ML competition champion.

## PROFESSIONAL EXPERIENCE

### Ford Motor Company

Sep 2026 – Present

#### Senior Data Scientist, Manufacturing Intelligence, Machine Learning & AI

- Architect graph network ontology of automotive manufacturing plant equipment within Neo4j and Google Cloud Platform.

Stack: Python, GCP, Neo4j, Docker, Make

### Caterpillar Inc.

Sep 2022 – Jun 2026

#### Lead Data Scientist

- Designed and deployed a TensorFlow ordinal deep-learning model predicting equipment repair times, replacing a regression model ( $R^2$  0.54) and using sentence transformer embeddings of labor descriptions across 2M+ combinations of labor code, sales model, and build variant. Reached 98% exact-class accuracy on the holdout set, tripled warranty coverage, and cut \$4M annually in time studies.
- Engineered the training labels rather than inheriting them: found warranty labor times were censored at the published standard, excluded them, and built ground truth from time studies, engineering estimates, and outlier-trimmed non-warranty medians.
- Governed the model with an explicit abstention rule (no prediction on unseen build variants or job codes) plus per-segment validation gates at a 90% floor; still delivered 95% coverage of warranty labor lines against an 80% business target.
- Owned the full MLOps lifecycle on AWS SageMaker (Infrastructure as Code, CI/CD, Docker, drift detection, and automated retraining) and architected the Snowflake data foundation behind the team's analytics platform.
- Delivered production ML across tabular, time series, and NLP applications for warranty adjudication, insurance reserving, and service quoting; mentored early-career data scientists on the team.

Stack: Python, R, SQL · PyTorch, TensorFlow/Keras, XGBoost, scikit-learn · Snowflake · AWS (SageMaker, Glue, Athena, Lambda, Step Functions, ECS/EKS, CloudWatch) · Docker · Azure DevOps, GitHub Actions

### Amazon.com Tempe, AZ

Feb 2022 – Sep 2022

#### Reliability Analytics Manager

- Replaced compliance-dependent failure reporting with a machine-generated signal: built a pipeline consuming hourly serialized component logs from S3 and reconstructed true component life from serial-number changes, then moved the fleet design standard from average life to B10.
- Applied Weibull and survival analysis across a global fleet of 400,000 autonomous robots, informing maintenance strategy and design metrics network-wide.
- Led a team of 5 BI engineers building a global robot monitoring pipeline (R/Shiny, AWS) adopted across sites for incident response and proactive maintenance.

Stack: R, Python · scikit-learn, AutoGluon · Shiny · PostgreSQL · Docker · AWS (Redshift, Athena, SageMaker, S3)

### Empire Cat Mesa, AZ

Jan 2012 – Feb 2022

#### Reliability Engineering Manager & Data Engineering Manager

Dec 2016 – Feb 2022

- Led modeling on a 25-person cross-functional program that replaced Caterpillar's estimated engine life target with an evidence-based one: reduced thousands of candidate factors via PCA to 9 controllable variables, then built a diesel engine digital twin on live telemetry to monitor them, extending haul truck engine life 23%.
- Connected reliability Weibull models to parts demand, which reduced part stockouts by 25%.
- Architected a component exchange program with data warehousing, ETL/ELT pipelines, and time series models to target at-risk components, with client-facing ownership of the reliability and product problem management programs.
- Established data engineering governance, quality standards, and a metadata framework adopted across the dealership.

#### Lead Reliability Engineer

Jun 2016 – Dec 2016

- Led root cause failure analysis on an aging mining truck line, remediating engine, drivetrain, and hydraulic failures to raise availability from 80% to 90%.
- Applied design of experiments and statistical modeling to isolate unreliability drivers; created an RMarkdown research framework Caterpillar incorporated into its CRM.

## Technical Communications Manager (Engineering Manager)

Dec 2012 – Jun 2016

- Engineering manager of a 5-person team delivering technical communications, reliability analytics, and data engineering for a 250-truck mining fleet.
- Applied DMAIC Six Sigma and data science to identify and prioritize reliability improvements across a fleet-wide program.
- Developed a fluids and wear analysis application (R, SQL) using oil tribology and spectroscopy to drive proactive maintenance, eliminating \$100K/year in license costs.

## Reliability Engineer

Jan 2012 – Dec 2012

- Led the Product Problem Management program, applying 8D and RCFA to resolve recurring failures and reduce downtime.
- Analyzed field failure data to prioritize the highest-impact failure modes and guide remediation planning.

Stack: SQL Server, T-SQL · R (ggplot2, tidyverse, RMarkdown) · Python (Pandas, scikit-learn) · SQLite · Jupyter · Docker · Ubuntu, Linux

## Road Machinery, LLC (now Komatsu) Phoenix, AZ

Jan 2008 – Jan 2012

## Technical Communicator (Engineer)

- Direct engineering support to clients and field service technicians, including root cause failure analysis, maintenance guidance, and life cycle cost analysis.
- Built the Total Cost of Ownership analysis underpinning a \$72M sale of electric mining haul trucks; principal consultant on a \$2.5M gold mine account in Mexico.

Stack: SQL Server, T-SQL · MATLAB/Octave · SolidWorks, finite element analysis · LaTeX

## Komatsu North America Chattanooga, TN

Jun 2006 – Dec 2007

## Design Engineer

- Led end-to-end development of a 25-ton hydraulic rescue winch from concept through prototyping, field testing, and release.
- Ran FMEA and FRACAS on prototype and production machines, authoring corrective actions that drove design changes and improved durability before release.

Stack: MATLAB/Octave · PTC · finite element analysis

## EARLY CAREER

Engineering Intern, Xtek Mining Services (now L&H Industrial) (2005) · Heavy Equipment Operator and Mechanic, Ryan Incorporated Central, Roland Machinery Co., and Harry W. Kuhn, Inc. (1997–2002), diagnosing and repairing mechanical, hydraulic, electrical, and control system failures on heavy capital earth moving equipment.

## EDUCATION

Bachelor of Science in Mechanical Engineering, Arizona State University

2006

Certificate of Apprenticeship in Operating Engineer, U.S. Department of Labor & I.U.O.E. Local 150 AISP

2001

## SELECTED CERTIFICATIONS

AWS Certified Machine Learning – Specialty (MLS-C01) · Lean Six Sigma Black Belt, Kennesaw State University · Certified SAFe Agilist · Advanced Deep Learning Specialist, IBM · Caterpillar Applied Failure Analysis

## RECOGNITION & SPEAKING

**Caterpillar CEO Award (2024):** AI-driven labor time model, for exceptional personal contribution and business impact.

**2x ML Competition Champion, Amazon (2022):** Internal regression and recommender system competitions.

**Invited Speaker, Data Science for Fleet Reliability (2021) at Caterpillar Six Sigma List Manager Workshop, Peoria, IL:** Diesel engine digital twin on live telemetry, tracking 9 PCA-derived controllable variables to set an evidence-based engine life target.

**Invited Speaker, Reliability & Rebuild Programs (2015, 2019) at Caterpillar, Tinaja Hills, AZ:** Product Problem Management; and a truck rebuild program at Caterpillar's large mining truck conference, delivering 2–4% higher availability at 60% of new-machine cost.

## CORE COMPETENCIES

**Machine Learning & AI:** Deep learning (PyTorch, TensorFlow/Keras) · time series forecasting · anomaly detection · survival analysis and censored data · ordinal regression · NLP and embeddings · design of experiments

**MLOps, Data & Cloud:** AWS SageMaker · Snowflake · CI/CD and Infrastructure as Code · Docker · model governance, validation gates, and drift detection · automated retraining · ETL/ELT pipelines · data architecture and governance · Linux

**Reliability Engineering:** Root cause failure analysis · 8D and 5 Whys · FMEA / DFMEA / PFMEA · FRACAS · Weibull analysis · prognostics and remaining useful life · condition monitoring · oil tribology and spectroscopy · life cycle cost analysis

**Domain Expertise:** Heavy equipment, mining, automotive and manufacturing · fleet and maintenance management · warranty and service operations · supply chain and demand planning